

SBSS Literature Review Comparison (Papers 1-10)

Paper 1

No.: 1

Title: Energy-Efficient Clustered Wireless Sensor Networks for Border Surveillance

Authors: Malathy et al.

Year: 2024

Thought (comparison):

Malathy and colleagues investigate energy-efficient clustered wireless sensor network (WSN) architectures for border surveillance, integrating ant colony optimization (ACO) algorithms for optimal routing performance. Their work addresses the fundamental challenge of balancing computational efficiency, energy consumption, and data security in resource-constrained environments requiring continuous operation. By organizing sensor nodes into hierarchical clusters with ACO-based routing and ECC cryptography, they achieve extended network lifetime and high packet delivery ratios validated through simulation.

In contrast, our proposed work implements direct WiFi bot-to-bot communication with real-time sensing and vision capabilities, avoiding complex multi-hop routing protocols entirely. Unlike their static clustered WSN approach using ACO for network-level routing optimization, we enable mobile, decentralized coordination through autonomous ESP32-CAM units. We emphasize local edge processing on ESP32-CAM for immediate autonomous responses to detected threats, eliminating routing delays and infrastructure complexity. While their system focuses on optimizing data transmission paths across static sensor networks, our design prioritizes on-device intelligence and context-aware decision-making without requiring centralized coordination or complex routing algorithms.

Match Factor: 35%

Paper 2

No.: 2

Title: Integrated Multi-Sensor Data Fusion for Border Control

Authors: Essendorfer et al.

Year: 2024

Thought (comparison):

Essendorfer and colleagues present an integrated framework for border surveillance emphasizing heterogeneous sensor data fusion and transnational information sharing through the SOBCAH project. Their work addresses the complexity of modern border security where infrared cameras, electro-optical systems, radar, and sonar must operate cohesively across vast terrain. The core innovation lies in a unified data fusion architecture integrating disparate sensor streams into a coherent operational picture through probabilistic reasoning, with shared databases enabling real-time information exchange among border agencies across multiple countries.

We emphasize local, real-time decision-making at the edge using ESP32-CAM for immediate autonomous responses, eliminating reliance on centralized databases and cross-border data sharing protocols used in their centralized architecture. Unlike their approach requiring complex multi-sensor fusion infrastructure and transnational coordination, our system performs human-animal classification directly on-device, enabling independent operation without external server dependencies. While their work excels at large-scale interoperability and multi-agency coordination, our design prioritizes autonomous edge intelligence, privacy preservation through local processing, and simplified deployment suitable for remote border regions lacking complex infrastructure. Our ESP32-CAM units can operate completely offline, making decisions locally without requiring the extensive communication infrastructure their system depends upon.

Match Factor: 30%

Paper 3

No.: 3

Title: Swarming Control for Unmanned Vehicles in Surveillance

Authors: Sauter et al.

Year: 2024

Thought (comparison):

Sauter and colleagues demonstrate swarming control algorithms for unmanned ground and aerial vehicles applied to surveillance and infrastructure protection. Their digital pheromone-based approach enables autonomous, adaptive coordination of heterogeneous sensor platforms, supporting complex behaviors including target tracking and multi-sensor fusion. The system mimics biological swarm intelligence where individual agents follow simple rules to achieve sophisticated collective behavior without centralized control, demonstrating robustness against individual agent failures and scalability to large deployments through real-world tests with UAVs, UGVs, and sensor networks.

We simplify swarm coordination into a practical, low-cost dual-bot system using direct WiFi communication, suitable for embedded deployment without the computational complexity of their digital-pheromone-based approach. Unlike their sophisticated swarm algorithms requiring complex coordination mechanisms and substantial computational resources, our ESP32-CAM based system implements straightforward bot-to-bot communication protocols optimized for resource-constrained edge devices. While their work demonstrates advanced multi-agent coordination with digital pheromones and complex behavioral algorithms, we prioritize deployment simplicity, cost-effectiveness, and immediate practicality for border surveillance. Our approach

trades some coordination sophistication for dramatically reduced system complexity, lower unit costs, and easier field deployment, making it more accessible for resource-limited border security applications.

Match Factor: 40%

Paper 4

No.: 4

Title: IoT-Based Smart Border Security with Machine Learning

Authors: Fatima et al.

Year: 2024

Thought (comparison):

Fatima and colleagues propose an IoT-based smart border security system utilizing machine learning for intruder detection, employing sensors, thermal imaging, and camera data analyzed by Faster R-CNN and EfficientDet models. Their architecture integrates edge computing for real-time processing with cloud-encrypted storage for long-term data retention and analysis, with automatic SMS alerts notifying security personnel upon threat detection. The system addresses practical deployment challenges in rugged terrains with limited manpower through multi-stage verification to mitigate false alarms and cloud integration for scalable remote monitoring.

We run lightweight CNN directly on ESP32-CAM through TinyML for offline operation with low latency, unlike their cloud-based ML models requiring constant connectivity and thermal imaging infrastructure. While their system depends on cloud resources for running computationally intensive models like Faster R-CNN and EfficientDet, our approach implements a custom lightweight neural network optimized specifically for ESP32-CAM's limited resources, enabling complete autonomous operation without internet connectivity. Unlike their requirement for thermal imaging cameras and cloud infrastructure, we achieve human-animal differentiation using standard RGB cameras with on-device inference, drastically reducing deployment costs and infrastructure requirements. Our system operates effectively in remote areas without cellular coverage or internet access, addressing a critical limitation in their cloud-dependent architecture.

Match Factor: 55%

Paper 5

No.: 5

Title: Wireless Sensor Networks for Border Surveillance

Authors: Arjun et al.

Year: 2024

Thought (comparison):

Arjun and colleagues survey multiple wireless sensor network (WSN) techniques for border surveillance and intruder detection, highlighting hybrid systems integrating seismic, acoustic, infrared, and camera sensors to detect human intrusions in diverse border scenarios. The paper discusses multi-layered architectures and communication protocols critical for large-scale deployments, examining how different sensor modalities complement each other through multi-sensor fusion approaches. They analyze challenges including false alarms from wildlife, sensor coverage limitations, and energy constraints, proposing intelligent data aggregation and energy-efficient communication protocols to extend network lifetime.

We move beyond theoretical discussion by building and validating a working multi-sensor (PIR, ultrasonic) and vision-based prototype demonstrating practical implementation challenges and solutions. Unlike their comprehensive survey which primarily discusses theoretical architectures and proposed approaches, our work provides a functional, tested system addressing real-world deployment constraints. While they examine various sensor fusion methodologies abstractly, we implement practical sensor integration on resource-constrained ESP32-CAM hardware, demonstrating how lightweight CNN models can effectively process camera data alongside PIR and ultrasonic sensors for human-animal differentiation. Our contribution includes actual hardware implementation details, power consumption measurements, and field-tested performance metrics rather than simulation results or theoretical frameworks, filling the gap between their surveyed concepts and deployable border surveillance solutions.

Match Factor: 45%

Paper 6

No.: 6

Title: Machine Learning Approaches Survey

Authors: Thomas et al.

Year: 2024

Thought (comparison):

Thomas and colleagues present a comprehensive survey on machine learning approaches, categorizing them into supervised, unsupervised, semi-supervised learning, and reinforcement learning methods. The paper explains key algorithms including decision trees, rule-based classifiers, naive Bayes, k-nearest neighbors, neural networks, and support vector machines, providing foundational understanding of their operational principles. They discuss applications across domains like pattern recognition, natural language processing, computer vision, and autonomous systems, offering insights directly relevant for AI-driven border surveillance and detection systems.

We apply ML concepts specifically by implementing CNN-based TinyML model on resource-constrained ESP32-CAM hardware for real-time visual intrusion detection, moving from theory to application. Unlike their broad theoretical survey covering multiple machine learning paradigms, our work focuses on practical deployment of convolutional neural networks optimized for edge devices with severe computational and

memory constraints. While they discuss various algorithms academically, we address the engineering challenges of quantizing, optimizing, and deploying neural networks on microcontrollers with less than 8MB RAM. Our contribution demonstrates how theoretical ML concepts translate into functioning edge-AI surveillance systems, including model compression techniques, inference optimization strategies, and real-time performance under resource constraints—practical aspects their survey does not explore.

Match Factor: 30%

Paper 7

No.: 7

Title: YOLO-Based Real-Time Border Security

Authors: Nahata et al.

Year: 2024

Thought (comparison):

Nahata and colleagues present a YOLO-based real-time border security surveillance system integrating deep learning with camera and UAV inputs to detect humans, vehicles, drones, and weapons. The paper highlights challenges in traditional border monitoring including limited visibility, vast coverage areas, and manual patrol inefficiencies. Their automated object detection approach using YOLO (You Only Look Once) architecture enables timely threat recognition with improved detection speed and accuracy, supporting integration with IoT devices for extended coverage and providing integrated alert mechanisms for rapid notification.

We replace YOLO with custom lightweight CNN optimized for ESP32-CAM's limited resources, trading minimal accuracy for drastically lower power consumption and cost, enabling edge processing. Unlike their approach requiring substantial computational resources to run YOLO models (typically on GPUs or high-end embedded platforms like Jetson boards), our lightweight CNN operates directly on ESP32-CAM's modest microcontroller. While YOLO-based systems excel in detection accuracy and multi-class object recognition, they require continuous power supply and expensive hardware unsuitable for distributed, solar-powered border deployments. Our design prioritizes energy efficiency, extended autonomous operation, and ultra-low unit cost (under \$10 per node) over detection sophistication, making large-scale deployment economically feasible. This trade-off enables deploying hundreds of units for comprehensive coverage at the cost of a few YOLO-based systems.

Match Factor: 60%

Paper 8

No.: 8

Title: Autonomous Robot Swarm for Border Intrusion

Authors: Mahjoub et al.

Year: 2024

Thought (comparison):

Mahjoub and colleagues develop an autonomous wireless sensor network employing a swarm of mobile robots for border intrusion detection, mimicking natural swarm behavior through distributed coordination algorithms. Their heterogeneous robot system collaboratively navigates complex terrains using YOLOv5 for object identification, with ROS (Robot Operating System) and Gazebo simulation environment utilized for system design, testing, and coordination protocol development. The study emphasizes cooperative navigation strategies, robust multi-robot communication protocols, and fault tolerance mechanisms enabling continued operation when subset of robots malfunction.

We adopt simpler, non-ROS architecture with low-power WiFi communication, prioritizing ease of deployment and cost-effectiveness over complex swarm behaviors using YOLOv5 and ROS. Unlike their sophisticated robotic platform requiring ROS middleware, complex simulation environments, and computationally intensive YOLOv5 models, our ESP32-CAM based system operates with minimal software dependencies and straightforward programming. While their approach demonstrates advanced multi-robot coordination and simulation-validated behaviors, it requires significant technical expertise for deployment and maintenance. Our design eliminates ROS complexity, replacing YOLOv5 with lightweight CNN suitable for microcontroller execution, and implements simple yet effective WiFi-based coordination that non-specialist personnel can deploy and maintain. This architectural simplification reduces development time, lowers technical barriers, and enables rapid field deployment without extensive simulation or complex coordination protocols.

Match Factor: 50%

Paper 9

No.: 9

Title: Mini Seeker Robot for Border Surveillance

Authors: Reddy et al.

Year: 2024

Thought (comparison):

Reddy and colleagues propose the Mini Seeker Robot, an automated border surveillance platform equipped with ultrasonic sensors, facial recognition, and live video streaming functionalities. Controlled via Raspberry Pi microcontroller and integrated with OpenCV computer vision library, the robot detects obstacles autonomously and identifies individuals by matching facial data with pre-registered databases. The cost-effective solution operates autonomously following pre-programmed patrol routes or under remote manual control, with ultrasonic sensors enabling collision avoidance and facial recognition providing identity verification capabilities.

We substitute Raspberry Pi with ESP32-CAM paired with lightweight ML, achieving significant reduction in unit cost, power draw, and physical footprint for wider field deployment. Unlike their Raspberry Pi-based platform which consumes approximately 2.5-4W during operation and costs \$35-75 per unit (depending on model), our ESP32-CAM system operates at under 1W and costs approximately \$8-12 per unit. While Raspberry Pi provides greater computational flexibility for running OpenCV and complex facial recognition algorithms, it requires substantially more power, making solar-powered deployment challenging. Our approach trades facial recognition capability for human-animal classification—a more relevant distinction for border surveillance—while achieving dramatically longer battery life and lower deployment costs. This enables deploying 5-10 ESP32-CAM units for every Raspberry Pi robot, providing superior coverage density and system redundancy.

Match Factor: 65%

Paper 10

No.: 10

Title: Helmet Detection and Number Plate Recognition

Authors: Sathe et al.

Year: 2024

Thought (comparison):

Sathe and colleagues focus on helmet detection and number plate recognition using deep learning models YOLOv5 and EasyOCR within traffic surveillance contexts. Their methodology involves bounding box overlap techniques to associate detected helmets with corresponding motorcyclists, effectively identifying helmet violation instances for traffic law enforcement. The study achieves high accuracy through transfer learning, leveraging pre-trained models and fine-tuning on domain-specific datasets, with comprehensive dataset augmentation including rotation, scaling, brightness adjustment, and synthetic variations improving model robustness.

We repurpose real-time object classification technology for identifying security threats (humans vs. animals) in border surveillance, adapting traffic monitoring techniques to security applications. Unlike their focus on traffic law enforcement through helmet and license plate detection, our system applies similar deep learning principles to border security contexts requiring human-animal differentiation. While their work demonstrates effective use of YOLOv5 and transfer learning for specific traffic scenarios, we adapt these methodologies to embedded edge devices with severe resource constraints. Their approach requires substantial computational resources for real-time video processing with YOLOv5 and OCR, whereas we implement simplified classification models optimized for ESP32-CAM's limited memory and processing capabilities. Our contribution demonstrates how traffic surveillance computer vision techniques can be efficiently adapted and compressed for deployment on ultra-low-cost edge devices in remote border monitoring applications.

Match Factor: 35%

Paper 11

No.: 11

Title: Digital Border Surveillance System using YOLOv7

Authors: Noor et al.

Year: 2024

Thought (comparison):

Noor and colleagues propose a Digital Border Surveillance System (DBSS) using YOLOv7 for real-time object detection to monitor illegal migration and trafficking across borders. The system integrates IoT sensors including PIR and ultrasonic sensors with a custom dataset of 7347 images featuring uniformed personnel, civilians, and animals for comprehensive detection capability. Achieving 89% alarm success rate and 85% notification accuracy, the system implements intelligent threat-level differentiation with severe alarms for civilians, moderate alerts for unknown uniforms, and minor notifications for animals, with lateral movement filtering and centralized control room dashboard for video logging and real-time monitoring.

We perform human-animal differentiation locally on ESP32-CAM, enabling immediate threat assessment without server communication latency used in their YOLOv7 server-side processing. Unlike their centralized architecture where YOLOv7 detection occurs on powerful backend servers requiring continuous network connectivity, our lightweight CNN executes directly on-device, eliminating communication delays and enabling autonomous operation in network-denied environments. While their system achieves impressive accuracy with YOLOv7's sophisticated detection capabilities, it requires substantial infrastructure including centralized control rooms and reliable network connections. Our edge-based approach provides sub-second threat assessment and classification without external dependencies, critical for remote border areas lacking reliable connectivity. This architectural difference enables our system to make autonomous decisions immediately upon detection rather than waiting for server-side processing and response commands.

Match Factor: 65%

Paper 12

No.: 12

Title: Thermal Image Pedestrian Detection

Authors: Kumar et al.

Year: 2024

Thought (comparison):

Kumar and colleagues develop a deep learning framework using YOLOv5 variants for pedestrian detection in thermal images, addressing limitations of RGB-based systems in challenging weather conditions including fog, rain, and nighttime operations. Trained on the UNIRI-TID thermal imaging dataset containing 7412 annotated images, YOLOv5-S achieves the highest mean Average Precision of 0.9908, outperforming tYOLOv3 across various thermal benchmarks while requiring less GPU and CPU resources. The study examines effects of layer freezing and data augmentation strategies, demonstrating particular relevance for border surveillance in deserts and mountainous regions where thermal imaging reliably detects human heat signatures regardless of lighting conditions.

We use standard RGB cameras with ultrasonic sensor fusion and lightweight ML, creating lower-cost solution optimized for daytime surveillance instead of thermal imaging for low-visibility conditions. Unlike their thermal imaging approach requiring specialized cameras costing \$500-5000+ per unit, our ESP32-CAM uses standard RGB sensors available for under \$10, dramatically reducing deployment costs. While thermal cameras excel in low-light and adverse weather conditions, they significantly increase system complexity and power requirements. Our design prioritizes cost-effectiveness and simplicity, acknowledging the trade-off of reduced nighttime performance for dramatically improved affordability and ease of deployment. For border surveillance applications with multiple detection points, deploying 50-100 RGB-based ESP32-CAM units provides superior daytime coverage density compared to a handful of thermal imaging systems, with supplementary infrared LEDs added to critical locations for nighttime enhancement at minimal cost.

Match Factor: 45%

Paper 13

No.: 13

Title: YOLO-Based Surveillance with Automated Alerts

Authors: Taneja et al.

Year: 2024

Thought (comparison):

Taneja and colleagues present a YOLO-based real-time surveillance system detecting persons, vehicles, weapons, and drones, replacing labor-intensive patrols with automated threat identification and email alerts via Flask web framework integration. The methodology employs dataset annotation using Roboflow platform, enabling efficient labeling and preprocessing of training data. YOLO training provides single-stage detection with superior speed over two-stage methods like Faster R-CNN, with YOLOv8 achieving 94.9% drone detection accuracy and 70% overall mAP across multiple object classes. The system implements OAuth2-secured email notifications and discusses future expansion including IoT deployment for distributed sensor networks.

We implement direct bot-to-command-center WiFi alerts functioning in isolated networks without internet dependency, unlike their email alerts requiring internet connectivity. While their Flask-based system with OAuth2-secured email notifications demonstrates sophisticated cloud integration, it fundamentally requires

internet connectivity and email infrastructure to function. Our ESP32-CAM units communicate directly via WiFi to local command stations using lightweight TCP/IP or MQTT protocols, enabling operation in remote border regions without cellular coverage or internet access. Unlike their reliance on cloud email services which introduce latency and infrastructure dependencies, our local alert system provides immediate notifications to nearby operators through dedicated WiFi networks or mesh configurations. This architectural choice prioritizes operational independence and zero-latency alerts over sophisticated cloud-based notification systems, making our solution viable in truly isolated deployment scenarios where internet connectivity is unavailable or unreliable.

Match Factor: 55%

Paper 14

No.: 14

Title: IoT-Based Solar Charger for Field Deployments

Authors: Daou et al.

Year: 2024

Thought (comparison):

Daou and colleagues utilize Arduino UNO R4 WiFi and WebSocket communication to a Flutter mobile app for real-time photovoltaic and battery parameter visualization, creating an IoT-based MPPT solar charger. They achieve over 97% tracking efficiency with closely matching laboratory and app readings by combining perturb-and-observe MPPT algorithm, high-precision sensing, and low-latency WebSocket telemetry. This demonstrates how tightly coupled embedded control, sensing, and real-time IoT communication can be integrated into single low-cost platforms, with direct application to field-deployed, power-aware surveillance assets in remote border regions where solar charging extends operational lifetime without frequent battery replacement.

We adapt WebSocket-based telemetry approach for transmitting surveillance bot status, sensor readings, and alerts to command center, repurposing solar monitoring technique for security applications. Unlike their focus on solar charging optimization and energy parameter monitoring, our system extends the IoT telemetry concept to transmit surveillance-relevant data including detection events, camera status, battery levels, and GPS coordinates. While their work demonstrates efficient solar power management crucial for remote deployments, we integrate similar real-time communication protocols for operational monitoring rather than just power metrics. Our ESP32-CAM implementation benefits from their demonstrated WebSocket approach for low-latency, bidirectional communication, adapting it to transmit surveillance alerts and receive configuration updates. This technology transfer from renewable energy monitoring to security surveillance demonstrates how IoT telemetry frameworks can be repurposed across domains, with our contribution being the practical integration of power management monitoring alongside threat detection capabilities in a unified edge-intelligent platform.

Match Factor: 35%

Paper 15

No.: 15

Title: 60 GHz CMOS-SOI Transceiver for IoT

Authors: Saponara et al.

Year: 2024

Thought (comparison):

Saponara and colleagues design essential components of a fully integrated 60 GHz CMOS-SOI transceiver for IoT nodes requiring multi-Gbps links over approximately 10 meters, targeting applications including vehicle-borne sensing systems, smart mobility, and surveillance. To meet ECMA-387-class link specifications at low power and compact area, they co-optimize wideband low-noise amplifiers, power amplifiers, and on-chip antennas, exploiting the 57-66 GHz unlicensed band. The millimeter-wave frequency enables high bandwidth for streaming multiple video feeds and sensor data simultaneously, though continuous power consumption must be carefully managed in battery-powered deployments, with short-range high-speed links ideal for vehicle-to-infrastructure communication.

We use standard WiFi offering longer range and lower power consumption suitable for surveillance applications, instead of high-speed, short-range 60 GHz transceivers for dense IoT networks. Unlike their millimeter-wave approach optimized for multi-Gbps data rates over 10-meter ranges, our 2.4 GHz WiFi implementation provides 50-100 meter range at modest data rates (1-10 Mbps) sufficient for surveillance video and sensor telemetry. While 60 GHz transceivers excel in bandwidth and interference immunity, they require line-of-sight conditions and consume significant power for RF operation. Our design prioritizes range, obstacle penetration, and power efficiency over bandwidth, recognizing that border surveillance applications require coverage area rather than ultra-high-speed links. Standard WiFi also benefits from ubiquitous infrastructure support, simplified debugging, and compatibility with existing network equipment, whereas 60 GHz systems require specialized hardware and expertise. This trade-off aligns with our emphasis on practical deployability and cost-effectiveness.

Match Factor: 25%

Paper 16

No.: 16

Title: IoT-Powered Surveillance Robots

Authors: Geetha M et al.

Year: 2024

Thought (comparison):

Geetha M and colleagues present mobile robot platforms for surveillance integrating IoT connectivity with onboard sensing for remote monitoring applications. The IoT-Powered Smart Surveillance Sentinel Bot emphasizes basic autonomy in relatively straightforward environments with standard communication models, while the Model Spy Bot introduces Li-Fi (Light Fidelity) technology for low-RF-emission, light-based communication enabling short-range covert surveillance. Li-Fi's directional nature provides inherent security against interception compared to omnidirectional radio communications, though trading transmission range and robustness for reduced detectability and electromagnetic interference immunity. These studies demonstrate how low-cost robotics and alternative physical-layer links can enhance surveillance coverage in specific operational contexts.

We integrate separate robot concepts into unified WiFi-based dual-bot system with coordinated sensor-vision fusion for comprehensive situational awareness. Unlike their presentation of two distinct robot platforms with different communication technologies (standard RF and Li-Fi), our system implements a cohesive architecture where multiple ESP32-CAM units coordinate via standard WiFi for practical deployment. While Li-Fi offers interesting covert communication properties, its requirement for line-of-sight and limited range make it impractical for outdoor border surveillance where units may be separated by hundreds of meters with terrain obstacles. Our WiFi-based approach provides reliable omnidirectional communication suitable for distributed surveillance networks while achieving adequate security through WPA2/WPA3 encryption. Rather than exploring alternative physical layers, we focus on optimizing standard WiFi for multi-bot coordination, implementing efficient protocols for sharing detection events and coordinating patrol coverage across multiple autonomous units.

Match Factor: 40%

Paper 17

No.: 17

Title: Navigation Interface for Multiple Robots

Authors: Sergio Saponara et al.

Year: 2024

Thought (comparison):

Sergio Saponara and colleagues develop a user-centric navigation interface and control framework enabling grouping and coordinated motion of multiple robots in shared environments. By providing interaction modalities and algorithms for grouping, tasking, and conflict management across multiple platforms, it reduces operator workload in multi-robot missions where single operators must manage numerous autonomous agents simultaneously. The interface allows operators to define robot groups, assign collective objectives, and monitor mission progress through intuitive visualization, with conflict management algorithms preventing collisions and deadlocks when multiple robots operate in proximity. Although primarily demonstrated in indoor or structured environments, the concepts directly apply to border-surveillance UGV swarms and patrol coordination.

We implement simplified, dedicated LCD-based command-center display focused on live border status monitoring rather than intricate navigation control interfaces. Unlike their sophisticated GUI framework designed for detailed robot navigation control, group management, and conflict resolution visualization, our command interface prioritizes essential surveillance information: detection events, bot status, battery levels, and coverage maps. While their system enables complex operator-directed navigation and task assignment, our design emphasizes autonomous operation with minimal operator intervention, reducing interface complexity to status monitoring and alert acknowledgment. Their approach suits applications requiring precise human-directed robot positioning, whereas border surveillance benefits more from autonomous patrol with exception-based human oversight. Our simplified LCD dashboard displays real-time detection feeds, threat classifications, and system health without navigation joysticks or complex tasking interfaces, reflecting our architectural philosophy of autonomous edge intelligence rather than centralized operator control.

Match Factor: 35%

Paper 18

No.: 18

Title: AI-Based Human Detection for Storage Optimization

Authors: Alajrami et al.

Year: 2024

Thought (comparison):

Alajrami and colleagues design an AI-based desktop program using OpenCV and Haar-cascade classifiers under Linux to trigger recording only when human body or face is detected, optimizing storage utilization. They report approximately 99% body-detection and 96% face-detection accuracy on test sequences by switching from motion-only triggering to human-specific detection, reducing video storage by 27% compared to continuous recording while maintaining complete coverage of human activity. The system greatly reduces search effort and storage costs by eliminating hours of empty footage from archived recordings, providing practical storage optimization for large-scale surveillance deployments.

We employ deep-learning-based CNN classifier for improved accuracy and robustness to lighting variations, moving beyond traditional Haar-cascade classifiers. Unlike their use of Haar-cascade classifiers—a classical computer vision technique effective but sensitive to lighting conditions, pose variations, and partial occlusions—our CNN-based approach learns hierarchical feature representations providing superior generalization. While Haar-cascades require careful manual feature engineering and struggle with non-frontal views or varying illumination, our lightweight CNN trained on diverse datasets handles lighting variations, different poses, and cluttered backgrounds more robustly. Their reported 99% body detection likely represents favorable test conditions, whereas CNN models maintain high accuracy across diverse real-world scenarios including shadows, backlighting, and partial occlusions common in outdoor border surveillance. Additionally, our CNN simultaneously performs human-animal classification rather than just human detection, providing actionable intelligence beyond their binary detection approach. This enhanced classification capability enables context-aware responses critical for border security applications.

Match Factor: 50%

Paper 19

No.: 19

Title: AI-Based Human Identification for Selective Recording

Authors: Eman Alajrami et al.

Year: 2024

Thought (comparison):

Eman Alajrami and colleagues discuss deployment factors for AI-based human identification including user acceptance, frame rate and resolution configuration, and integration with existing CCTV infrastructure. Evaluations with real organizations demonstrate that people find AI-assisted selective recording both useful and easy to use, suggesting successful integration into existing systems without requiring complete camera replacement. The study examines practical considerations including computational requirements, network bandwidth, storage architecture, and operator training needs, with user acceptance testing revealing that security personnel readily adopt AI-assisted systems when they demonstrably reduce workload without compromising security coverage.

We implement mobile robotic surveillance platform that actively patrols and scans areas rather than passively monitoring fixed camera feeds with AI triggering. Unlike their focus on retrofitting existing fixed CCTV infrastructure with AI-based selective recording, our system represents a fundamentally different paradigm: mobile autonomous surveillance rather than static camera upgrades. While their work addresses the practical challenge of enhancing legacy surveillance systems through software upgrades, we design purpose-built mobile platforms integrating sensing, intelligence, and mobility from the ground up. Their approach serves building security and facility monitoring where camera positions are predetermined, whereas border surveillance requires dynamic coverage of vast areas impossible to monitor with fixed cameras alone. Our mobile ESP32-CAM bots provide flexible deployment, adaptive coverage patterns, and the ability to investigate detected anomalies by moving closer, capabilities unavailable to fixed cameras regardless of AI enhancements. This represents a complementary rather than competing approach to surveillance system design.

Match Factor: 30%

Paper 20

No.: 20

Title: Next-Generation Defense Monitoring with Digital Twin, Blockchain, and AI

Authors: Geetha et al.

Year: 2024

Thought (comparison):

Geetha and colleagues propose a next-generation defense monitoring system incorporating digital twin concepts, blockchain, IoT, and AI-based facial recognition for secure asset tracking and access control. The system provides immutable logs, automated responses via smart contracts, and high resilience to tampering by utilizing streaming face detection/recognition, GPS-based activity logging, and permissioned blockchain (Hyperledger Fabric). Experimental results in military-style facilities demonstrate strong resistance to simulated cyberattacks, approximately 98% face-recognition accuracy under favorable conditions, GPS localization within about 5 meters, and blockchain logging latency of 2-3 seconds. The blockchain architecture eliminates single points of failure with distributed ledger ensuring data integrity even if individual nodes are compromised.

We prioritize real-time intrusion detection and threat classification over individual identification and access management using blockchain and facial recognition. Unlike their sophisticated system combining digital twins, blockchain immutability, and facial recognition for facility access control and asset tracking, our design focuses on rapid perimeter intrusion detection with human-animal classification rather than individual identity verification. While their blockchain-based approach ensures tamper-proof audit trails and smart-contract-driven access control valuable for high-security facilities, it introduces computational overhead and latency unsuitable for time-critical border intrusion alerts. Our system trades comprehensive identity management and blockchain auditability for millisecond-level threat detection and classification, recognizing that border surveillance primarily requires knowing what crossed the border (human vs. animal) rather than who. Their 2-3 second blockchain latency, though impressive, exceeds acceptable response times for autonomous intrusion alerts where immediate action is critical.

Match Factor: 25%

Paper 21

No.: 21

Title: Blockchain for Defense Applications

Authors: Mohith S et al.

Year: 2024

Thought (comparison):

Mohith S and colleagues examine how blockchain enhances auditability and decentralization in defense monitoring systems, removing single points of failure inherent in traditional password and ID systems stored in centralized databases. The distributed ledger provides tamper-evident logs of all access attempts, movements, and security events, creating comprehensive audit trails for forensic analysis and compliance verification. Unlike conventional systems where database compromise exposes all credentials, blockchain's cryptographic linking and distributed storage make retroactive modification computationally infeasible. However, the paper highlights ongoing challenges with blockchain scalability and latency for very large deployments, particularly

in real-time defense operations requiring immediate decision-making, emphasizing need for lightweight consensus mechanisms and careful architectural partitioning.

We focus on operational performance of real-time surveillance rather than implementing ledger-based authentication and audit trails using blockchain. Unlike their emphasis on tamper-proof audit logging and decentralized credential management through blockchain technology, our system prioritizes millisecond-level threat detection and immediate autonomous response capabilities. While blockchain provides valuable security properties for audit trails and access control, its inherent latency (even with optimized consensus) conflicts with our requirement for instantaneous edge-based decision-making. Our ESP32-CAM units must detect, classify, and respond to intrusions within hundreds of milliseconds—timeframes incompatible with blockchain consensus overhead. We accept conventional security measures and local logging in exchange for zero-latency autonomous operation, recognizing that border surveillance success depends primarily on detection speed rather than cryptographically guaranteed audit immutability. For applications requiring forensic audit trails, our system can forward logs to blockchain-enabled backend systems post-detection without impacting real-time operational performance.

Match Factor: 20%

Paper 22

No.: 22

Title: Privacy-Preserving Border Surveillance

Authors: Ramyashree P M et al.

Year: 2024

Thought (comparison):

Ramyashree P M and colleagues note that conventional border-surveillance architectures frequently emphasize detection performance through cameras, ground sensors, and radar while placing little emphasis on formal privacy guarantees and secure data handling. The paper examines cryptographic primitives including encryption, secure aggregation, and privacy-preserving analytics used in surveillance and IoT streams, alongside sophisticated deep learning-based human and object detection techniques. Proposing a border surveillance architecture combining deep learning-based target detection with cryptographic methods to safeguard transmitted and stored data, the system employs secure aggregation allowing multiple sensor nodes to collectively compute detection statistics without revealing individual sensor data, and homomorphic encryption enabling computation on encrypted data without decryption.

We prioritize low-latency local processing and rapid alerting, avoiding computational overhead of heavy cryptographic operations on live data streams. Unlike their cryptographically intensive approach implementing homomorphic encryption and secure multi-party computation for privacy-preserving analytics, our edge-processing architecture achieves privacy through data locality—sensitive visual data never leaves the device. While their system encrypts and securely aggregates data across network nodes, we eliminate transmission of raw video entirely, sending only classification results (human/animal detection events) with optional encrypted

thumbnails. Their cryptographic approach suits scenarios requiring centralized analytics while maintaining privacy, whereas our design questions the need for centralized processing altogether. By performing all sensitive computations on-device, we achieve inherent privacy without cryptographic overhead that would exceed ESP32-CAM's computational capacity. This architectural approach provides practical privacy through edge intelligence rather than computationally expensive privacy-preserving cryptography.

Match Factor: 35%

Paper 23

No.: 23

Title: AI-Powered Search and Rescue Rover

Authors: Dilbar Roshan et al.

Year: 2024

Thought (comparison):

Dilbar Roshan and colleagues present a human detection rover for AI-powered search and rescue in disaster zones, integrating PIR motion detection with YOLO V3 object recognition and GPS tracking for locating survivors. Their system uses Raspberry Pi microcontroller with Wi-Fi/MQTT communication enabling wireless remote control and cloud-based monitoring, achieving 96% detection accuracy across various environments and functioning effectively even in cluttered or low-visibility conditions. PIR sensors provide initial motion alerts triggering camera-based YOLO detection for human verification, reducing false positives from animal movement or environmental factors, with GPS coordinates of detected survivors transmitted immediately to rescue coordination centers. Future enhancements include thermal imaging and SLAM integration for autonomous navigation in complex environments.

We adapt sensor-vision fusion technology specifically for border security, differentiating between human threats and non-threatening animals, repurposing disaster response technology. Unlike their disaster response application focused on locating any human survivors in rubble or hazardous areas, our border surveillance system must distinguish between authorized/unauthorized humans and wildlife to generate appropriate responses. While their YOLO V3 implementation on Raspberry Pi prioritizes detection sensitivity to find any human presence, our lightweight CNN on ESP32-CAM implements classification distinguishing humans from animals—a critical capability for reducing false alarms from wildlife in border regions. Their system appropriately treats all human detections equally, whereas border security requires threat assessment beyond mere presence detection. We repurpose their sensor fusion concept (PIR + vision) but optimize for different operational requirements: continuous autonomous monitoring with animal filtering rather than survivor location in disaster scenarios. This represents technology transfer across domains with significant adaptation for different mission-critical requirements.

Match Factor: 45%

Paper 24

No.: 24

Title: Hybrid Bot Detection Framework

Authors: Greeshma Lingam et al.

Year: 2024

Thought (comparison):

Greeshma Lingam and colleagues propose a hybrid bot-detection framework combining reinforcement learning and swarm-optimization concepts to improve identification accuracy in large social networks. Rather than depending solely on static rules, the model continuously learns behavioral cues from user interactions and reshapes decision boundaries over time through adaptive learning mechanisms. The authors highlight the system's adaptive nature helping identify rapidly evolving bot patterns that evade traditional detection methods, with reinforcement learning enabling the system to improve detection strategies based on feedback from confirmed bot identifications and false positive corrections. Swarm optimization distributes detection across multiple agents that share information about suspicious patterns, with findings showing improved precision and reduced false positives compared to rule-based or single-model approaches.

We apply ML to physical security domain—classifying real-world intruders at borders rather than identifying automated accounts in cyberspace. Unlike their application of machine learning to detect automated social media accounts through behavioral pattern analysis, our system applies computer vision and edge AI to physical border surveillance for detecting and classifying tangible threats. While their hybrid framework analyzes digital user behavior metrics, interaction patterns, and account characteristics in cyberspace, we process visual imagery and sensor data from physical environments to identify humans and animals. The learning paradigms differ fundamentally: they employ reinforcement learning to adapt bot detection strategies based on social network feedback, whereas we use supervised learning with labeled training data of humans and animals for classification. Despite both systems employing adaptive ML approaches, the domains (cyber vs. physical), input modalities (behavioral data vs. visual/sensor), and threat types (fake accounts vs. border intrusions) differ completely, making direct technical comparison limited to high-level AI concepts rather than implementation details.

Match Factor: 15%

Paper 25

No.: 25

Title: Portable GPS Tracking Solution

Authors: Sudhanshu Janwadkar et al.

Year: 2024

Thought (comparison):

Sudhanshu Janwadkar and colleagues present a portable GPS tracking solution designed for real-time monitoring of individuals in outdoor locations, retrieving accurate positional coordinates and transmitting them wirelessly to remote base stations for applications including child safety, elderly assistance, and asset tracking. Through compact hardware design combining GPS receiver, microcontroller, and wireless transmitter, the system provides straightforward, cost-effective field-level tracking capability with basic mapping functions visualizing tracked positions on digital maps. The authors demonstrate reliable performance under varied outdoor conditions including urban, rural, and partially forested environments, with power optimization techniques extending battery life for multi-day autonomous operation and geofencing capabilities generating alerts when tracked entities enter or exit predefined boundary zones.

We integrate GPS with environmental sensing and AI vision for comprehensive threat assessment beyond mere positional tracking. Unlike their focus on passive location monitoring of known tracked entities (children, elderly, assets), our system combines GPS positioning with active environmental sensing (PIR, ultrasonic) and AI-powered visual threat detection for autonomous border surveillance. While their GPS tracking provides valuable position data, it lacks situational awareness—knowing what is happening at that location. Our ESP32-CAM units augment positional information with intelligent context: detecting human vs. animal presence, assessing movement patterns, and capturing visual evidence. Their geofencing capability triggers location-based alerts, whereas our system generates threat-classified alerts with visual confirmation. We extend basic GPS tracking into an intelligent surveillance node providing not just "where" but "what, who, and what type of threat," transforming simple position monitoring into actionable security intelligence suitable for autonomous border patrol operations.

Match Factor: 40%

Paper 26

No.: 26

Title: FPGA-Based Rescue Bot

Authors: K. Sudhakar et al.

Year: 2024

Thought (comparison):

K. Sudhakar and colleagues develop an FPGA-based rescue bot for surveillance combining internet-controlled mobility with PIR sensor-based human detection and camera-based video transmission for disaster response and border monitoring applications. The robot integrates distance sensors for obstacle detection and environmental monitoring units for hazard assessment, utilizing FPGA's parallel processing capabilities to handle multiple sensor streams simultaneously without delay. FPGA implementations provide deterministic timing and hardware-level concurrency unavailable in sequential microcontroller architectures, enabling guaranteed real-time performance with reconfigurability supporting field updates and mission-specific customization. The authors highlight such architectures enhancing onsite safety by allowing robots to enter areas humans cannot safely access.

We opt for microcontroller-based (ESP32) architecture offering lower-cost, power-efficient solution while maintaining adequate performance, instead of FPGA for parallel processing. Unlike their FPGA implementation providing superior parallel processing and deterministic real-time guarantees, our ESP32 microcontroller approach dramatically reduces system cost (ESP32-CAM: \$8-12 vs. FPGA development boards: \$100-500+) and power consumption while sacrificing some processing capability. FPGAs excel at parallel sensor fusion and guaranteed latency but require specialized programming expertise (Verilog/VHDL), longer development cycles, and higher power consumption. Our microcontroller-based design accepts sequential processing with adequate performance for border surveillance requirements while enabling rapid development using standard C/C++ programming, simplified debugging, and wider developer accessibility. For applications requiring absolute deterministic timing or massive parallelism, FPGAs are superior; however, border surveillance tolerates modest processing latency making microcontroller architecture more cost-effective. We prioritize deployment economics and developer accessibility over maximum parallel processing capability.

Match Factor: 50%

Paper 27

No.: 27

Title: Laser Beam Intrusion Detection

Authors: Anisom Chakraborty et al.

Year: 2024

Thought (comparison):

Anisom Chakraborty and colleagues propose a laser beam intrusion detection mechanism for securing outdoor perimeters across long distances, where any interruption to the laser line triggers instant alerts through an IoT platform allowing remote monitoring personnel to assess situations quickly. With high precision and low power consumption, the system is suitable for continuous deployment along fence lines and boundary perimeters, reducing false alarms compared to complex radar or camera-based perimeter systems as laser beam interruption provides unambiguous intrusion indication. Multiple parallel laser beams at different heights can detect crawling, walking, or climbing intrusion attempts, with IoT connectivity enabling centralized monitoring of extensive perimeters from single control rooms with geo-located alerts directing response teams to specific breach points.

We deploy mobile bots with active scanning capabilities creating dynamic surveillance zones rather than static laser beam perimeter defenses. Unlike their fixed laser-beam approach creating static detection lines along predetermined perimeters, our mobile ESP32-CAM bots provide dynamic, adaptive surveillance coverage. While laser beams excel at detecting perimeter breaches with zero false positives from environmental factors, they only detect crossing events at fixed locations without providing threat classification or visual confirmation. Our mobile platform combines detection with classification (human vs. animal) and visual documentation while offering flexible deployment—units can patrol, reposition, or concentrate on high-risk areas as threats evolve. Laser systems require clear line-of-sight along entire perimeter and are vulnerable to deliberate circumvention (crawling under, jumping over configured beam heights), whereas our mobile vision-based approach provides

comprehensive area coverage with intelligent threat assessment. We trade laser beam's simplicity and precision for mobile surveillance's flexibility, adaptability, and contextual intelligence.

Match Factor: 30%

Paper 28

No.: 28

Title: Multi-Sensor Building Monitoring

Authors: Tao Qi et al.

Year: 2024

Thought (comparison):

Tao Qi and colleagues propose a building monitoring system combining readings from multiple sensors to reduce false alarms and improve decision reliability. By incorporating temperature, gas, motion, and environmental sensors and processing them collectively, the system filters noise that would confuse single-sensor setups, with wireless connectivity for remote supervision and timely alert delivery. Results show multi-sensor fusion significantly improves accuracy by requiring corroboration across sensor modalities before generating alerts, drastically reducing false positives from individual sensor malfunctions or environmental anomalies. The fusion logic implements probabilistic reasoning weighing sensor reliability and consistency across time to distinguish genuine threats from spurious readings, making the system dependable for security-critical applications including border checkpoints, guard posts, and command facilities.

We adapt multi-sensor fusion concept combining PIR and ultrasonic sensors with AI vision specifically for border intrusion monitoring and false alarm reduction. Unlike their building monitoring application integrating temperature, gas, and environmental sensors for facility safety (fire, chemical leaks, structural issues), our system fuses motion detection (PIR), proximity sensing (ultrasonic), and visual classification (CNN) specifically for human-animal differentiation in outdoor border environments. While their fusion logic addresses building-specific threats through environmental monitoring, we optimize sensor combination for distinguishing security-relevant targets from wildlife and environmental motion. Their probabilistic reasoning approach for sensor corroboration directly informs our multi-sensor design, where PIR triggers camera activation, ultrasonic provides range confirmation, and CNN delivers threat classification—layered verification reducing false alarms from wind-blown vegetation, small animals, or environmental factors. We repurpose their fusion principles from indoor facility monitoring to outdoor border surveillance with sensor selection and fusion algorithms adapted for different operational contexts and threat types.

Match Factor: 50%

Paper 29

No.: 29

Title: Autonomous IoT Surveillance Robot

Authors: Muhamad Syazmie Sepeeh et al.

Year: 2024

Thought (comparison):

Muhamad Syazmie Sepeeh and colleagues develop an autonomous robot integrating IoT connectivity and onboard sensing to facilitate remote monitoring. With basic navigation and obstacle detection capabilities, the robot patrols designated areas while streaming sensor updates to a user interface accessible via web browsers or mobile applications, demonstrating potential for combining mobility with IoT communication for small-scale surveillance in indoor facilities or limited outdoor perimeters. Ultrasonic or infrared sensors enable collision avoidance during autonomous patrol following pre-programmed paths or waypoints, with operators able to manually intervene to redirect robots or investigate specific areas remotely. The system is capable of handling larger areas or more complex environments with additional hardware enhancements including more sophisticated sensors and advanced navigation algorithms.

We enhance basic IoT patrol robot with ML-based threat classification and active, radar-style ultrasonic scanning for intelligent surveillance behavior. Unlike their foundational IoT robot providing basic patrol and remote monitoring capabilities, our system integrates on-device machine learning for autonomous threat assessment without operator interpretation. While their robot streams sensor data requiring human operators to identify significant events, our ESP32-CAM performs intelligent classification on-device, alerting operators only to confirmed human detections while filtering animal activity autonomously. Their basic ultrasonic obstacle avoidance focuses on collision prevention, whereas we implement active scanning patterns creating virtual radar coverage detecting and ranging potential threats. We extend their proof-of-concept IoT surveillance robot into an intelligent autonomous system capable of making operational decisions independently—distinguishing threats from non-threats and generating prioritized alerts. This progression from teleoperated sensing platform to autonomous intelligent surveillance represents significant functional enhancement beyond basic IoT connectivity and remote control.

Match Factor: 55%

Paper 30

No.: 30

Title: ESP32-CAM Surveillance Robot

Authors: Gurpreet Kaur et al.

Year: 2024

Thought (comparison):

Gurpreet Kaur and colleagues present a compact surveillance robot using ESP32-CAM module to enable live video transmission over Wi-Fi networks, remotely controlled via web interface providing real-time visual

feedback for basic inspection and monitoring tasks. The authors demonstrate the low-cost nature of the design, showing that effective surveillance does not always require expensive components or complex systems, with ESP32's integrated Wi-Fi capability eliminating need for separate communication modules. Camera resolution and frame rate are adjustable to balance video quality against network bandwidth constraints, with simple motor control allowing forward, backward, and turning movements directed through web-based joystick interface. The work demonstrates how readily available microcontroller development boards can be transformed into functional robotic systems for indoor and outdoor observation, with limitations including relatively short Wi-Fi range and basic obstacle avoidance.

We extend ESP32-CAM platform by adding AI-powered human-animal differentiation and integrating feedback mechanisms with command center for coordinated response. Unlike their basic teleoperation system streaming raw video for human operator interpretation, our implementation adds edge-based machine learning enabling autonomous threat classification on-device. While their web-controlled robot requires continuous operator attention to identify significant events in video streams, our system autonomously detects, classifies, and alerts operators only to relevant threats (humans) while filtering routine animal activity. We enhance their simple motor control with intelligent behavior patterns including automated scanning, patrol routes, and threat investigation protocols. Their web-joystick interface requires dedicated operator control, whereas our command-center integration enables one operator supervising multiple autonomous units—drastically improving operational efficiency. We build upon their foundational ESP32-CAM platform, transforming it from remotely operated camera into an intelligent autonomous surveillance node capable of independent operation while maintaining centralized coordination for multi-unit deployments.

Match Factor: 70%

Status: Papers 1-30 completed **Remaining:** Papers 31-50 **Ready for:** Continuation to papers 31-40